

Hackathon

Problem scope: Predictive healthcare analysis to assess the likelihood of patients developing medical conditions using data-driven insights, enabling early and preventive care.

Primary Users: Patients

Problem Statement

Patients often remain unaware of their health risks until medical conditions become severe and symptomatic. Although large volumes of patient health data such as medical history, lifestyle factors, and clinical measurements are generated over time, this information is rarely used to proactively assess individual health risks.

Current healthcare systems primarily follow a reactive approach, where patients seek care only after experiencing noticeable symptoms. Existing risk assessment methods are often manual, generalized, or inaccessible to patients, making it difficult for individuals to understand their personal risk levels or take timely preventive action.

As a result, many patients miss critical opportunities for early intervention and preventive care, leading to delayed diagnosis, increased treatment costs, and poorer health outcomes. There is a need for a scalable, data-driven healthcare analytics solution that empowers patients with early risk insights and supports proactive health decision-making.

Core Value:

The system empowers patients to assess their current health risk by analyzing their historical medical records and providing a clear risk classification **low, medium, or high** to support early awareness and proactive health decisions.

What does the system do??

- Analyzes **historical and recent patient health data**, including vital signs and basic clinical information
- Detects **patterns and trends over time** that indicate potential patient deterioration
- Predicts a **patient deterioration risk level** and classifies it as **Low, Medium, or High**
- Provides **clear, easy-to-understand risk insights** to patients for early awareness
- Supports **proactive health monitoring and preventive decision-making**

System Architecture

1 Inputs (What the system receives)

- Patient medical records
- Vital signs (HR, BP, SpO₂, etc.)
- Basic clinical info (age, conditions)
- Time-based data (trends over hours/days)

2 Processing (What the system does internally)

- Data preprocessing & normalization
- Trend detection over time
- ML-based risk scoring
- Conversion of score → Low / Medium / High
- Explainability layer (why this risk?)

3 Outputs (What the patient sees)

- Clear risk category (Low / Medium / High)
- Simple explanation of contributing factors
- Recommended **next actions** (monitor / consult / urgent attention)

Impact and Use Cases:

Impact:

The proposed patient deterioration risk system enables a shift from **reactive healthcare to proactive health management** by providing early risk awareness to patients.

Key impacts include:

- Early identification of patients at risk of health deterioration before critical events occur
- Improved patient awareness and involvement in managing their own health
- Timely medical attention for high-risk patients, potentially reducing emergency situations
- Support for preventive care, leading to better health outcomes and reduced long-term healthcare costs

This system empowers patients with actionable insights while supporting safer and more efficient healthcare delivery.

Use Cases:

1 Patient Health Monitoring

Patients can upload or connect their medical records and vital data to:

- Continuously assess their health risk level
- Understand changes in risk over time
- Take timely action based on risk classification

2 Remote & Preventive Healthcare Platforms

The system can be integrated into:

- Remote health monitoring solutions
- Preventive care applications
- Telehealth platforms

to identify early warning signs and prioritize patient follow-ups.

3 Hospital & Clinical Support (Secondary Use Case)

While patient-focused, the system can support healthcare providers by:

- Highlighting high-risk patients
- Assisting in patient prioritization
- Enabling proactive intervention planning

Future Scope:

The patient deterioration risk system can be expanded beyond its initial implementation to support broader and more advanced healthcare use cases.

Potential future enhancements include:

- Extension of risk assessment to cover **multiple health conditions** and patient profiles
- Integration with **wearable devices and IoT health sensors** to enable real-time risk monitoring
- Advanced **explainable AI features** to provide deeper insights into risk factors
- Personalized, data-driven health insights tailored to individual patient trends

Scalability:

The system is designed to scale efficiently across different levels of healthcare adoption:

- **Individual Level:** Personal health risk monitoring for patients
- **Organizational Level:** Integration with hospitals, clinics, and telehealth platforms
- **Population Level:** Aggregated analytics to support public health monitoring and preventive care initiatives

With cloud-based deployment, the solution can support large-scale data processing while maintaining reliability and accessibility.

